

Token-Level Entropy Reveals Demographic Disparities in Language Models

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Abstract

We ask whether demographic identity, signaled by a name alone, systematically reshapes the generative distribution of a language model. Measuring full-vocabulary Shannon entropy at temperature zero across six open-weight base models and 5,760 implicit sentence-completion prompts (e.g., “*Tanisha walked into the office on a Monday morning and*”), we find that Black-associated names produce *higher* first-token entropy than White-associated names across all six architectures — opposite to the output-level homogeneity bias documented under explicit demographic prompting [Lee et al., 2024] — and Black-associated names always produce greater entropy above identity-neutral baselines than White-associated names ($\Delta\Delta > 0$ in all six models). Women-associated names co-occur with *lower* first-token entropy (DL-pooled $\hat{\beta}_{\text{gender}} = -0.041$, $p = .019$) and more homogeneous outputs (DL-pooled $\hat{\alpha}_{\text{gender}} = +0.024$, $p < .001$) than men-associated names — a pattern convergent with homogeneity bias; race and gender effects are additive. Instruction tuning does not attenuate the race gap (matched-format DL-pooled $\hat{\beta} = +0.153$). Running the same templates with explicit group labels instead of names yields null race effects in 10 of 12 models where implicit probing is significant — establishing that probing methodology is a primary determinant of which distributional structure is recovered.

1 Introduction

Despite growing evidence of output-level demographic disparities — including homogeneity bias, the finding that models generate more uniform text about marginalized groups [Lee et al., 2024, Cheng et al., 2023] — the structure of the pretraining distribution itself remains poorly characterized. Temperature modulation fails to mitigate homogeneity bias. We ask directly: does demographic identity, signaled by a name alone, systematically reshape the generative distribution?

We measure full-vocabulary Shannon entropy at temperature zero across six open-weight base models and six instruction-tuned counterparts, using 5,760 sentence-completion prompts with implicit name-based probing (“*Tanisha walked into the office on a Monday morning and*”). All models receive raw-text prompts; instruct models are additionally probed under native chat-formatted input.

The race result runs counter to output-level homogeneity bias: Black-associated names produce *higher* first-token entropy than White-associated names — significant in five of six architectures, directionally positive in all six — and Black-name continuations are more semantically diverse in all six. Men-associated names produce higher first-token entropy than women-associated names (DL-pooled $\hat{\beta}_{\text{gender}} = -0.041$, $p = .019$; 4/6 significant), and women-associated name outputs are more homogeneous (DL-pooled $\hat{\alpha}_{\text{gender}} = +0.024$, $p < .001$) — a pattern convergent with homogeneity bias; race and gender effects are additive. Identity-neutral baselines reveal a consistent asymmetry:

Black-associated names produce greater entropy above the Someone/[PERSON] reference than White-associated names in all six models ($\Delta\Delta > 0$). Instruction tuning does not attenuate the race gap; a within-study explicit-label comparison yields null race effects in 10 of 12 models where implicit probing is significant.

Our contributions are: (1) the first systematic measurement of first-token entropy disparities across four intersectional groups using full-vocabulary probabilities and implicit name-based probing across six base architectures; (2) matched-format probing of instruction-tuned counterparts isolating training-stage from formatting effects; and (3) a within-study implicit/explicit probing comparison establishing that methodology determines which distributional structure is recovered.

2 Related Work

Homogeneity bias. Lee et al. [2024] first documented that ChatGPT generates more homogeneous narratives (by cosine similarity of sentence embeddings) for racial minorities and women than for dominant groups. The pattern extends to vision-language models [Lee and Jeon, 2025, 2024] and to LLM simulations that produce caricatures of marginalized identities [Cheng et al., 2023].

Name-based implicit probing. Using demographic names as implicit probes has established roots in audit research: resume-callback disparities [Bertrand and Mullainathan, 2004], word-embedding geometry [Caliskan et al., 2017], and occupational stereotyping in NLP [Lucy and Bamman, 2021, Navigli et al., 2023]. A key advantage over explicit prompting is ecological validity: the model responds to learned co-occurrence statistics rather than an explicit trigger that may activate alignment-induced suppression. We use Tzioumis [2018] to screen names for demographic signal strength.

Token-level entropy. Shannon entropy of the next-token distribution is used for hallucination detection [Farquhar et al., 2024, Kuhn et al., 2023] and calibration analysis [Wiher et al., 2022]. We extend these approaches to full-vocabulary entropy under implicit demographic probing, base models, and matched-format instruct comparisons.

3 Method

3.1 Demographic Groups and Names

We study four intersectional groups: Black women, Black men, White women, and White men. Thirty first names per group are drawn from audit literature [Bertrand and Mullainathan, 2004, Caliskan et al., 2017] and cross-referenced against Tzioumis [2018] to verify unambiguous single-race associations. For each name we record its tokenized length (`name_token_length`) under each model’s tokenizer; length is measured for the name in isolation (not embedded in the full prompt), using each model’s default tokenizer without special tokens. Two *baseline* conditions ("Someone", "[PERSON]") isolate template-level entropy and enable interpretation of group effects as deviations from a no-identity reference. The reported H_1^{baseline} is the mean first-token entropy pooled across all 96 baseline prompts (48 templates $\times$ 2 placeholder conditions); $\Delta H_1^{\text{group}}$ denotes the group mean H_1 minus this pooled baseline mean.

3.2 Prompt Templates

Each name is combined exhaustively with 48 sentence-completion templates, yielding $30 \times 48 = 1,440$ prompts per group and 5,760 total. Templates are neutral-scenario sentence fragments spanning everyday contexts (domestic, social, shopping, recreation, health, transportation, meals, neighborhood) without presupposing demographic characteristics. For example: “*Tanisha walked into the office on a Monday morning and*”. A complete listing is in Appendix D.

3.3 Models

We evaluate twelve decoder-only causal language models: six base checkpoints spanning 7–12B parameters and six instruction-tuned counterparts (Table 6 in Appendix A). All are run locally with `torch_dtype="auto"` and `device_map="auto"`.

Instruct models are probed under two conditions. First, the *matched-format* condition uses the same raw-text sentence-fragment inputs as base models, holding input format constant to isolate training-stage effects. Second, *native chat-formatted* probing wraps each prompt as a single user turn via `tokenizer.apply_chat_template()` with no explicit system message; for Qwen3-8B, thinking-mode tokens are suppressed via `enable_thinking=False`. This condition characterizes deployed-usage behavior.

3.4 Inference Protocol

Inference is run at temperature zero using HuggingFace `model.generate()` with `output_scores=True`, returning the full logit vector over the vocabulary at each step. We generate up to 100 tokens per prompt. Shannon entropy at step t is:

$$H_t = - \sum_{v \in \mathcal{V}} p_t(v) \log p_t(v) \quad (1)$$

where $\mathcal{V}$ is the full model vocabulary and $p_t(v)$ is the softmax probability of token v .

3.5 Metrics and Statistical Model

Primary metric. First-token entropy H_1 is maximally tied to the name itself, as no self-generated tokens have yet intervened.

Secondary metric. Mean-over-sequence entropy $\bar{H} = T^{-1} \sum_{t=1}^T H_t$ is reported for comparability. We fit a linear mixed-effects model per LLM:

$$H_{ij} = \beta_0 + \beta_1 \text{race}_i + \beta_2 \text{gender}_i + \beta_3 (\text{race} \times \text{gender})_i + \beta_4 \ell_i^c + u_i + \varepsilon_{ij} \quad (2)$$

where i indexes names, j indexes templates, ℓ_i^c is mean-centered name-tokenization length, and u_i is a name-level random intercept. Race is coded 0=White, 1=Black; gender is coded 0=Man, 1=Woman, so positive $\hat{\beta}_{\text{race}}$ indicates Black > White entropy and positive $\hat{\beta}_{\text{gender}}$ indicates Woman > Man entropy. When the name random-effect variance collapses to zero (singular fit), we use OLS with cluster-robust standard errors clustered by name. Singular fits were obtained for all six base models, all six instruct models under matched raw-text probing, and five of six instruct models under chat-formatted probing; the exception is Falcon3-10B-Instruct under chat-formatted probing, where the LME converged. All other reported estimates use OLS with cluster-robust standard errors. The LME specification above represents the intended model; OLS is used as the fallback when the name random-effect variance collapses to zero. A supplementary frequency-controlled model (+log-name-prevalence and template fixed effects) is reported in Appendix L.

4 Experiments

4.1 Setup

All 5,760 prompts are processed per model in batches of 8 with left-padded tokenization. Each JSONL record includes `group`, `name`, `name_token_length`, `template_id`, `first_token_entropy`, `mean_entropy`, and the full `per_token_entropy` sequence.

4.2 Base Model Results

4.2.1 Tokenization Audit

Mean name-tokenization lengths by group and model are reported in Appendix B (Table 7). The Black > White asymmetry is universal across all six models. Gemma-2-9B presents the largest tokenization asymmetry yet produces the smallest first-token race effect ($d = 0.008$) — a dissociation that constrains a pure tokenization-artifact interpretation. All subsequent regression estimates include ℓ_i^c as a covariate.

Model	ANOVA (H_1)		Race: Black–White		Gender: Woman–Man	
	$F(3, 5756)$	p	d [95% CI]	p	d [95% CI]	p
Qwen3-8B-Base	0.28	.839	+0.013 [$\pm .052$]	.405	+0.017 [$\pm .052$]	.527
Llama-3.1-8B	3.23	.021	+0.053 [$\pm .052$]	.013	−0.063 [$\pm .052$]	.021
Falcon3-10B	27.17	< .001	+0.181 [$\pm .052$]	< .001	−0.154 [$\pm .052$]	< .001
Mistral-NeMo-12B	2.76	.040	+0.060 [$\pm .052$]	.004	−0.046 [$\pm .052$]	.040
Gemma-2-9B	0.17	.913	+0.013 [$\pm .052$]	.472	−0.014 [$\pm .052$]	.472
OLMo-2-7B	26.49	< .001	+0.201 [$\pm .052$]	< .001	−0.119 [$\pm .052$]	< .001

Table 1: One-way ANOVA and pooled main-effect Cohen’s d for first-token entropy (H_1). Race: all Black minus all White prompts ($n = 2,880$ per pool). Positive race d : Black > White; negative gender d : Man > Woman.

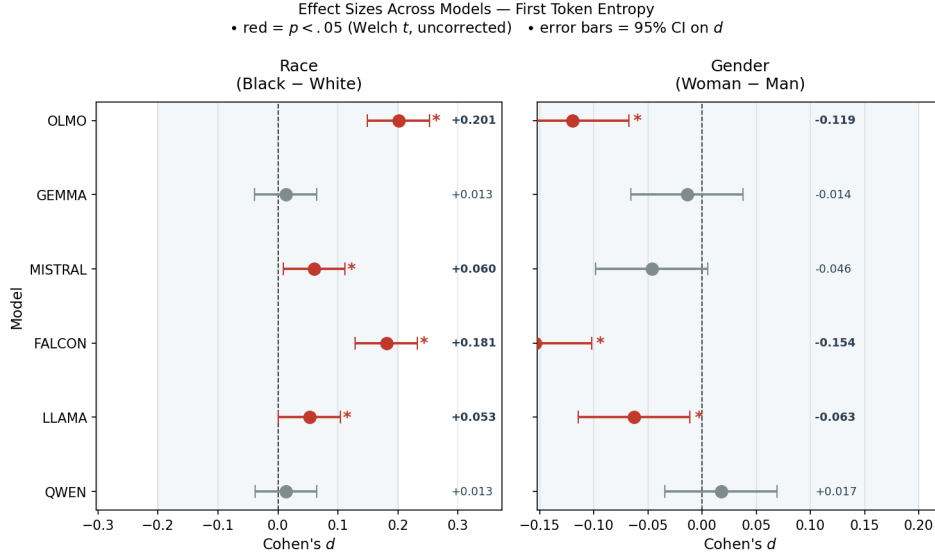


Figure 1: Forest plot of Cohen’s d for race (Black–White, left) and gender (Woman–Man, right) main effects on H_1 . Red dots: $p < .05$. Error bars: 95% CI. Shaded band: $|d| < 0.2$.

4.2.2 First-Token Entropy: Main Effects

Table 1 reports omnibus ANOVA and Cohen’s d for H_1 . The race effect is positive (Black > White) in all six models and significant in four by one-way ANOVA (Llama, Falcon, Mistral, OLMo). Figure 1 displays the forest plot. Group means and standard deviations are in Appendix F; full pairwise Welch t -tests (Bonferroni-corrected) are in Appendix E. The most consistent pairwise contrast is Black Man vs. White Woman, significant in 4 of 6 models.

The gender effect is negative (Man > Woman) in five of six models and significant in four by one-way ANOVA (Llama, Falcon, Mistral, OLMo). Falcon shows the largest effect ($d = -0.154$, $p < .001$), followed by OLMo ($d = -0.119$, $p < .001$), Llama ($d = -0.063$, $p = .021$), and Mistral ($d = -0.046$, $p = .040$). Qwen is the only model with a positive estimate ($d = +0.017$, $p = .527$); Gemma shows a small non-significant negative effect ($d = -0.014$, $p = .472$).

4.2.3 Mean-Over-Sequence Entropy

Entropy by sequence position (Figure 2, Appendix G) shows group separation strongest at position 1, decaying within 10–20 tokens as self-generated context accumulates. Accordingly, mean-entropy race effects are weaker: significant positive effects in Falcon3-10B and OLMo-2-7B, a reversed effect in Mistral-NeMo-12B (White > Black mean entropy). Gender mean-entropy effects follow the same direction as H_1 : men > women in five of six models, significant in two (Falcon, OLMo). Full mean-entropy regression results are in Appendix G.

Model	H_1^{baseline}	$\Delta H_1^{\text{Black}}$	$\Delta H_1^{\text{White}}$	$\Delta\Delta$
Qwen3-8B-Base	3.814	+0.087	+0.072	+0.015
Llama-3.1-8B	4.316	+0.123	+0.072	+0.051
Falcon3-10B	4.022	+0.228	+0.033	+0.194
Mistral-NeMo-12B	4.341	+0.123	+0.066	+0.057
Gemma-2-9B	3.488	-0.031	-0.042	+0.011
OLMo-2-7B	4.158	+0.042	-0.139	+0.181

Table 2: First-token entropy relative to identity-neutral baselines (Someone/[PERSON]). $\Delta\Delta > 0$ in all six models: Black-associated names produce greater entropy above baseline than White-associated names.

4.2.4 Baseline Conditions

Table 2 reports first-token entropy relative to identity-neutral baselines (Someone/[PERSON]). The differential $\Delta\Delta = \Delta H_1^{\text{Black}} - \Delta H_1^{\text{White}}$ is positive in all six models: Black-associated names always produce greater entropy above the no-identity reference than White-associated names. Black-associated names produce entropy above baseline ($\Delta H_1^{\text{Black}} > 0$) in five of six models. Entropy for White-associated names is heterogeneous across models in both sign and magnitude — near-zero in Falcon (+0.033 nats), moderately positive in Qwen, Llama, and Mistral (+0.066–+0.072 nats), and negative in Gemma and OLMo (−0.042 and −0.139 nats respectively).

The consistent finding is therefore not about the absolute position of White-associated names relative to the baseline, which varies substantially, but about the gap: Black-associated names are associated with greater distributional departure from the no-identity reference than White-associated names in every model tested.

A parallel pattern holds for gender. In five of six models, men-associated names produce greater entropy above the identity-neutral baseline than women-associated names ($\Delta H_1^{\text{Man}} > \Delta H_1^{\text{Woman}}$): Llama (+0.128 vs. +0.067 nats), Falcon (+0.213 vs. +0.048), Mistral (+0.116 vs. +0.073), Gemma (−0.030 vs. −0.042), and OLMo (+0.005 vs. −0.103). Qwen is the only exception (+0.070 vs. +0.089). Women-associated names thus sit closer to the no-identity reference distribution than men-associated names in five of six architectures — the gender-level analogue of the race asymmetry, and convergent with the lower first-token entropy co-occurring with women-associated names throughout the main analysis.

4.2.5 Mixed-Effects Model Results

Table 3 reports fixed-effect estimates from Equation 2 with name-tokenization length controlled. The race effect is statistically significant in five of six models and directionally positive in all six; Qwen3-8B-Base is the exception ($\hat{\beta} = +0.050$, $p = .238$), with a near-zero primary-level effect ($d = 0.013$) that does not reach significance even under tokenization control. Gemma-2-9B similarly shows $d = 0.013$ at the primary level ($p = .087$) yet reaches significance under tokenization control ($\hat{\beta} = +0.028$, $p = .003$); this inflation is consistent with name tokenization length as a statistical suppressor in Gemma — longer-tokenized Black names co-occur with modestly lower raw H_1 values, and the tokenization covariate accounts for this association, reducing its attenuation of the observed race coefficient. R^2 values are below 1.5%, indicating demographic group explains a small fraction of overall entropy variance while the disparity is robust in direction.

Cross-model consistency. All six $\hat{\beta}_{\text{race}}$ estimates are positive. Because the models share overlapping corpora, we use two dependence-robust cross-model tests. A DerSimonian–Laird random-effects meta-analysis [DerSimonian and Laird, 1986] yields pooled $\hat{\beta}_{\text{race}} = +0.096$, 95% CI [+0.045, +0.147], $p < .001$ ($I^2 = 84.4\%$). The harmonic mean p -value [Wilson, 2019], valid under arbitrary dependence, is $p_{\text{HMP}} = 4 \times 10^{-6}$. The binomial sign test (all six positive) yields $p = .016$. The substantial I^2 confirms that effect magnitude varies considerably across architectures while the direction is consistent.

Gender effects. Men-associated names co-occur with higher first-token entropy than women-associated names in five of six models, significant in four (Llama, Falcon, Mistral, OLMo; Table 3).

Model	Race effect ($\hat{\beta}_{\text{race}}$, Black–White)				Gender ($\hat{\beta}_{\text{gender}}$, W–M)			R^2
	$\hat{\beta}$	95% CI	p	sig	$\hat{\beta}$	95% CI	p	
Qwen3-8B-Base	+0.050	[−0.033, +0.132]	.238		+0.039	[−0.014, +0.092]	.149	< .001
Llama-3.1-8B	+0.079	[+0.027, +0.131]	.003	**	−0.044	[−0.079, −0.009]	.015	< .001
Falcon3-10B	+0.255	[+0.152, +0.359]	< .001	***	−0.156	[−0.235, −0.078]	< .001	.015
Mistral-NeMo-12B	+0.061	[+0.025, +0.097]	.001	***	−0.040	[−0.068, −0.012]	.005	< .001
Gemma-2-9B	+0.028	[+0.009, +0.047]	.003	**	−0.010	[−0.021, +0.001]	.086	< .001
OLMo-2-7B	+0.189	[+0.104, +0.275]	< .001	***	−0.098	[−0.168, −0.027]	.006	.014

Table 3: OLS regression estimates (cluster-robust SEs, clustered by name) for H_1 , controlling for name tokenization length. $n = 5,760$ obs, 120 name clusters. Interaction $\hat{\beta}_{\text{race} \times \text{gender}}$ is non-significant in all models (all $p > .15$) and omitted. ** $p < .01$, *** $p < .001$.

Model	Race: Black–White			Gender: Woman–Man		
	$\hat{\alpha}_{\text{race}}$	p	sig	$\hat{\alpha}_{\text{gender}}$	p	sig
Qwen3-8B-Base	−0.036	.003	**	+0.001	.911	
Llama-3.1-8B	−0.066	< .001	***	+0.028	.011	*
Falcon3-10B	−0.037	.005	**	+0.029	.023	*
Mistral-NeMo-12B	−0.057	< .001	***	+0.052	< .001	***
Gemma-2-9B	−0.059	< .001	***	+0.009	.484	
OLMo-2-7B	−0.039	< .001	***	+0.022	.019	*

Table 4: LME MPCs coefficients with template random intercept. Negative race $\hat{\alpha}$: Black-name outputs are more diverse; positive gender $\hat{\alpha}$: women-name outputs are more homogeneous. * $p < .05$, ** $p < .01$, *** $p < .001$.

DL random-effects pooling yields $\hat{\beta}_{\text{gender}} = -0.041$ (95% CI [−0.075, −0.007], $p = .019$; $I^2 = 82.1\%$). Falcon shows the largest estimate in magnitude ($\hat{\beta} = -0.156$, $p < .001$); Qwen is the only positive estimate ($\hat{\beta} = +0.039$, $p = .149$). The substantial I^2 confirms that effect magnitude varies considerably across architectures — as it does for the race effect — while the direction is consistent in five of six models. Unlike the race finding, the gender direction is convergent with homogeneity bias [Lee et al., 2024]: women-associated names are associated with a more constrained generative distribution, consistent with prior evidence of greater output homogeneity for women under explicit prompting. The race $\times$ gender interaction is non-significant in all six models (all $p > .15$; Table 3 caption), indicating that the gender and race effects are additive and do not moderate each other.

4.3 Output-Level Homogeneity

We embed every generated continuation with Sentence-BERT (all-mpnet-base-v2; Reimers and Gurevych 2019), masking the generating name, and compute *mean pairwise cosine similarity* (MPCS) per (template, group) cell. Each cell contains the 30 continuations generated from the 30 names within that demographic group for a given template; MPCs measures how similar those 30 outputs are to one another, with lower values indicating more diverse outputs. Group membership is thus used to define the comparison set at the output stage, while prompts themselves use implicit name-based probing throughout. We fit a per-LLM LME with template as a random intercept (Equation 3; implementation details in Appendix J):

$$\text{MPCS}_{tk} = \alpha_0 + \alpha_1 \text{race}_k + \alpha_2 \text{gender}_k + \alpha_3 (\text{race} \times \text{gender})_k + u_t + \varepsilon_{tk} \quad (3)$$

The race coefficient is negative and significant in all six models (Table 4): Black-associated name outputs are *more* semantically diverse — not more homogeneous — than White-associated name outputs. This is the opposite of homogeneity bias documented under explicit prompting [Lee et al., 2024], but is convergent with the entropy results. The gender coefficient is positive in all six models and significant in four (Llama, Falcon, Mistral, OLMo): women-associated name outputs are more homogeneous than men-associated name outputs (DL-pooled $\hat{\alpha}_{\text{gender}} = +0.024$, 95% CI [+0.010, +0.037], $p < .001$; $I^2 = 52.5\%$; sign test $p = .016$). This is directionally convergent with the first-token entropy finding: women-associated names co-occur with both lower H_1 and more

homogeneous output sequences, while men-associated names co-occur with higher H_1 and more diverse outputs. A within-prompt analysis of the H_1 -diversity link is reported in Appendix K.

4.4 Instruction-Tuned Model Results

4.4.1 Native Chat-Template Probing

One-way ANOVA and Cohen’s d for native chat-template probing are reported in Table 20 (Appendix O). The pattern is substantially more heterogeneous than base-model results. After tokenization control (Table 21), only Llama-3.1-8B-Instruct shows a significant positive race effect ($\hat{\beta} = +0.039$, $p = .019$); Mistral’s large negative primary-level effect ($d = -0.161$) is absorbed by tokenization control, consistent with a tokenization-length artifact. Gender effects on H_1 are also more heterogeneous than base-model results: the tokenization-controlled coefficient is negative in three models and positive in three, with only Qwen ($\hat{\beta} = -0.023$, $p = .009$) and OLMo ($\hat{\beta} = -0.144$, $p = .017$) reaching significance. The directional consistency of the base-model gender effect (5/6 negative) does not replicate under chat-template formatting.

4.4.2 Output-Level Homogeneity and Format Effect

Output-level diversity (MPCS) under chat-formatted probing is more robust than H_1 : 5/6 instruct models show significant negative race coefficients (Table 22), directionally consistent with base-model results. Gender effects on MPCS are also more stable than H_1 : women-associated outputs are more homogeneous (positive $\hat{\alpha}_{\text{gender}}$) in four of six instruct models, significant in two (Llama, Mistral), directionally consistent with base-model results.

The contrast between matched raw-text probing (Appendix N: all six instruct models positive and significant, DL-pooled $\hat{\beta} = +0.153$) and native chat-template probing (1/6 significant after tokenization control) identifies chat-template formatting as the factor most consistently associated with attenuation of the first-token demographic signal. The distributional race asymmetry is not attenuated by RLHF/SFT under matched-format probing; its attenuation at the first generation step co-occurs with chat-template input formatting. Output-level diversity (MPCS) is less sensitive to this format difference.

4.5 Within-Study Probing Comparison

To enable a direct within-study comparison of implicit and explicit probing on the same models, we collected output-diversity data under an *explicit-label* condition using the same 48 templates but substituting a demographic group label in the name slot (e.g., “A Black woman was in the middle of washing the dishes when”). Because only one prompt exists per (template, group) cell under explicit labeling, 30 completions per cell were collected via temperature sampling ($T = 1.0$). MPCS is computed identically to the implicit condition (Section 4.3, Appendix J).

The two conditions differ in two respects: the demographic signal is carried by an explicit group label rather than a name, and output diversity reflects stochastic variation across repeated samples of a single prompt rather than variation across 30 demographically distinct names at temperature zero. Table 5 reports the race $\hat{\alpha}$ under both conditions for all 12 models.

Implicit probing yields a significant negative race effect in 11 of 12 models. Explicit probing yields mostly null effects: significant in 2 of 12 models, and in both cases negative (consistent with implicit, not reversed). No model produces a significant positive race effect — Black outputs less diverse — under explicit group-label probing. Effect sizes are substantially attenuated in the explicit condition relative to implicit probing in every model.

This within-study comparison qualifies the cross-study contrast with Lee et al. [2024]. The divergence between probing methods is most precisely characterized as a signal/null contrast in these models: implicit name-based probing recovers a consistent output-diversity gap while explicit group-label probing does not, in 10 of 12 models. The homogeneity bias pattern — Black outputs less diverse under explicit prompting — documented in Lee et al. [2024] does not appear in these architectures under explicit probing; whether it is model-family specific or requires other prompt structures is left to future work. What the within-study comparison establishes directly is that probing methodology determines whether a race signal in output diversity is detectable, independently of its direction.

Model	Implicit (name-based, $T = 0$)		Explicit (group-label, $T = 1$)	
	$\hat{\alpha}_{\text{race}}$	p	$\hat{\alpha}_{\text{race}}$	p
<i>Base models</i>				
Qwen3-8B-Base	−0.036	.003	−0.006	.295
Llama-3.1-8B	−0.066	< .001	−0.018	.001**
Falcon3-10B	−0.037	.004	−0.004	.443
Mistral-NeMo-12B	−0.057	< .001	−0.008	.134
Gemma-2-9B	−0.059	< .001	−0.009	.089
OLMo-2-7B	−0.039	< .001	−0.009	.104
<i>Instruction-tuned (chat-formatted)</i>				
Qwen3-8B	−0.036	< .001	−0.014	.261
Llama-3.1-8B-I	−0.045	< .001	+0.009	.303
Falcon3-10B-I	−0.056	< .001	+0.001	.946
Mistral-NeMo-I	−0.013	.334	−0.028	.002**
Gemma-2-9B-IT	−0.044	< .001	+0.014	.238
OLMo-2-7B-I	−0.035	< .001	−0.007	.429

Table 5: Within-study comparison of MPCs race effects ($\hat{\alpha}_{\text{race}}$, Black−White) under implicit name-based and explicit group-label probing across all 12 models. Negative $\hat{\alpha}$: Black outputs more diverse. Implicit: significant in 11 of 12 models. Explicit: significant in 2 of 12 (both negative, same direction as implicit); no model shows a significant positive effect. ** $p < .01$.

5 Discussion

Black-associated names consistently co-occur with higher first-token entropy and more semantically diverse outputs than White-associated names across all six base architectures (significant in 5/6 under tokenization-controlled regression; $\Delta\Delta > 0$ against identity-neutral baselines in all six). This direction is opposite to output-level homogeneity bias, and it persists through instruction tuning under matched-format probing.

Asymmetric distance from the no-identity baseline. In all six models, Black-associated names produce greater entropy above the Someone/[PERSON] baseline than White-associated names ($\Delta\Delta > 0$; Table 2). Black-associated names produce entropy above baseline in five of six models; entropy for White-associated names is heterogeneous in sign and magnitude across models. The consistent finding is not the absolute position of White-associated names relative to the baseline, which varies substantially, but the gap: Black-associated names always produce greater entropy above the no-identity reference. A parallel asymmetry holds in five of six models, where men-associated names produce greater entropy above the identity-neutral baseline than women-associated names.

What the direction tells us. Higher entropy for Black-associated names indicates a less constrained next-token distribution. Two factors are consistent with this pattern: frequency-indexed sparsity (less prevalent names are associated with flatter distributions) and richer contextual associations (more diverse co-occurrence patterns). A sparsity decomposition (Appendix M) shows that frequency-indexed sparsity accounts for only $\approx 20\%$ of the observed gap in the two largest-effect models (Falcon3-10B, OLMo-2-7B). A template-category analysis shows the race gap is context-structured in all six models (larger in Routine and Social contexts, smaller in transactional contexts), inconsistent with uniformly flat priors from underrepresentation.

Instruction tuning and chat-template formatting. Matched-format probing of all six instruct models shows instruction tuning does not attenuate the first-token entropy race gap (DL-pooled $\hat{\beta} = +0.153$, exceeding the base-model estimate; Appendix N). The attenuation under native chat-formatted probing (1/6 significant) co-occurs with input formatting rather than training stage. Output-level diversity (MPCS) is less sensitive to this format difference, remaining significant in 5/6 instruct models under chat probing.

Divergence between implicit and explicit probing. Implicit name-based probing yields significant race effects in 11 of 12 models; explicit group-label probing on the same templates yields mostly null effects (2 of 12 significant; Table 5). The tools used to audit model bias substantially shape what

biases are detected: audits relying solely on explicit demographic framing may systematically miss the distributional structure recoverable through implicit probing.

Gender effects. Men-associated names co-occur with higher first-token entropy (DL-pooled $\hat{\beta}_{\text{gender}} = -0.041$, 95% CI $[-0.075, -0.007]$, $p = .019$; 4/6 significant, 5/6 directionally negative) and more diverse output sequences (DL-pooled $\hat{\alpha}_{\text{gender}} = +0.024$, 95% CI $[+0.010, +0.037]$, $p < .001$; 4/6 significant, 6/6 directionally positive) than women-associated names across base models. The two measures are convergent: women-associated names are associated with more constrained generative distributions at both the first-token and output level. Unlike the race effect, this direction is consistent with homogeneity bias [Lee et al., 2024] — women generating more homogeneous outputs than men under explicit prompting. The race $\times$ gender interaction is non-significant in all six models for both metrics, indicating that the gender and race effects are additive rather than moderated by each other. The gender effect is cross-architecturally variable in magnitude ($I^2 = 82.1\%$ for H_1) but directionally consistent, a pattern structurally parallel to the race finding.

Limitations. All analyses are observational; causal claims require experimental manipulation. Matched raw-text probing of instruct models is outside their designed input distribution and appropriate only for the training-stage question. First-token entropy integrates over the full vocabulary and reflects many factors; more targeted token-category analyses would be needed for finer-grained characterization. Eight cross-demographic names are flagged (sensitivity analysis in Appendix I); a White Man pool sensitivity (Appendix I.1) finds the race effect strengthens in 5/6 models when class-coded names are excluded. Results cover four intersectional groups in the Black/White racial binary, the 7–12B parameter range, and English-dominant models.

6 Conclusion

Across six open-weight base models and 5,760 implicit sentence-completion prompts, Black-associated names produce higher first-token entropy and more semantically diverse outputs than White-associated names in all six architectures — opposite to output-level homogeneity bias. Black-associated names always produce greater entropy above the no-identity reference than White-associated names ($\Delta\Delta > 0$). Men-associated names co-occur with higher first-token entropy and more diverse outputs than women-associated names — convergent with homogeneity bias — with race and gender effects additive. Instruction tuning does not attenuate the race gap under matched-format probing; attenuation under chat-formatted probing co-occurs with input formatting rather than training stage. Explicit group-label probing on the same templates yields null race effects in 10 of 12 models where implicit probing is significant: the tools used to audit model bias substantially shape what is detected.

Broader Impacts

This work has two concrete implications for model auditing practice. First, the divergence between implicit and explicit probing results indicates that audit methodology is not neutral: tools designed around explicit demographic framing may systematically miss or invert the distributional asymmetries present in the pretraining distribution. Second, the persistence of the race gap through instruction tuning — under matched-format probing — indicates that alignment training does not address pretraining-level distributional asymmetries, with implications for whether post-training is a sufficient response to representational disparities. The consistent asymmetry in distance from the no-identity baseline — Black-associated names always further from that reference than White-associated names — suggests that representational equity cannot be evaluated at the output level alone. Releasing code and data will support independent replication and extension to additional demographic groups, languages, and model families.

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A Models Evaluated

Key	Model	Params	Family	Checkpoint
<i>Base models</i>				
qwen	Qwen3-8B-Base [Qwen Team, 2025]	8B	Qwen	base
llama	Llama-3.1-8B [Grattafiori et al., 2024]	8B	Llama	base
falcon	Falcon3-10B-Base [TII, 2024]	10B	Falcon	base
mistral	Mistral-NeMo-12B [Mistral AI, 2024]	12B	Mistral	base
gemma	Gemma-2-9B [Gemma Team, 2024]	9B	Gemma	base
olmo	OLMo-2-7B [Groeneveld et al., 2024]	7B	OLMo	base
<i>Instruction-tuned counterparts</i>				
qwen_it	Qwen3-8B [Qwen Team, 2025]	8B	Qwen	instruct
llama_it	Llama-3.1-8B-Instruct [Grattafiori et al., 2024]	8B	Llama	instruct
falcon_it	Falcon3-10B-Instruct [TII, 2024]	10B	Falcon	instruct
mistral_it	Mistral-NeMo-Instruct-2407 [Mistral AI, 2024]	12B	Mistral	instruct
gemma_it	Gemma-2-9B-IT [Gemma Team, 2024]	9B	Gemma	instruct
olmo_it	OLMo-2-7B-Instruct [Groeneveld et al., 2024]	7B	OLMo	instruct

Table 6: Models evaluated. Base models are confirmed non-instruction-tuned checkpoints. Each instruct model is probed under two conditions: matched raw-text (identical to base-model inputs) and native chat-template formatting.

B Name Tokenization Lengths

Model	Black Woman	Black Man	White Woman	White Man
Qwen3-8B-Base	2.20	2.37	2.00	1.27
Llama-3.1-8B	2.17	2.37	2.00	1.27
Falcon3-10B	1.80	1.83	1.07	1.00
Mistral-NeMo-12B	2.23	2.27	2.10	1.43
Gemma-2-9B	1.73	1.60	1.00	1.00
OLMo-2-7B	2.20	2.37	2.00	1.27

Table 7: Mean name-tokenization length (subword tokens per name) by group and model. Black-associated names tokenize longer than White-associated names in all six models. Gemma-2-9B has one of the largest asymmetries yet the smallest first-token race effect, constraining a pure tokenization artifact explanation.

C Effect Size in Context

Race effects ($d = 0.013\text{--}0.201$; Table 1) are small by Cohen’s (1988) conventions but comparable to the foundational audit benchmark: the callback-rate disparity in Bertrand and Mullainathan [2004] corresponds to $d \approx 0.20\text{--}0.36$. Directional consistency across six base models — confirmed by DL meta-analysis and a dependence-robust harmonic mean p -value ($p_{\text{HMP}} = 4 \times 10^{-6}$) — is stronger evidence of a systematic pattern than any single effect size.

D Prompt Templates

The 48 sentence-completion templates are listed below. The {name} slot is replaced with a name from the corresponding group pool.

- (T1) {name} was in the middle of washing the dishes when
- (T2) {name} sat down on the couch after a long day and
- (T3) {name} opened the refrigerator and noticed that
- (T4) {name} was folding laundry when

(T5) {name} heard a noise outside the front door and
(T6) {name} turned off the lights before getting into bed and
(T7) {name} was making breakfast in the kitchen when
(T8) {name} looked around the living room and decided to
(T9) {name} ran into an old friend at the
(T10) {name} was on the phone with a friend when
(T11) {name} sat next to a stranger on the bus and
(T12) {name} was waiting outside for
(T13) {name} sent a text message that said
(T14) {name} waved to the neighbor across the street and
(T15) {name} stood in line at the grocery store and
(T16) {name} was looking at the menu and
(T17) {name} walked into the pharmacy and
(T18) {name} returned the item to the store and
(T19) {name} counted out the cash and
(T20) {name} laced up their shoes and headed out to
(T21) {name} sat down on a bench in the park and
(T22) {name} was in the middle of reading when
(T23) {name} turned on the television and
(T24) {name} signed up for the class at the
(T25) {name} arrived early at the community center and
(T26) {name} took a walk around the block and
(T27) {name} packed a bag for the weekend and
(T28) {name} called to make a doctor's appointment because
(T29) {name} handed the nurse the paperwork and
(T30) {name} picked up the prescription at the counter and
(T31) {name} decided to stay home that day because
(T32) {name} stepped onto the bus and
(T33) {name} stood at the bus stop while
(T34) {name} got into the car and
(T35) {name} arrived at the station just as
(T36) {name} locked the bike to the rack and
(T37) {name} picked up the phone and called
(T38) {name} received a message that said
(T39) {name} wrote a reminder on the notepad to
(T40) {name} set an alarm for the morning and
(T41) {name} checked the weather and
(T42) {name} cleared the table after dinner and
(T43) {name} ordered what they usually got and
(T44) {name} brought food to share and
(T45) {name} finished eating and pushed back the chair and
(T46) {name} noticed the flyer posted on the wall and
(T47) {name} stopped to chat with a neighbor about
(T48) {name} walked past the old building on the corner and

E Full Pairwise t -Test Results

Table 8 reports pairwise Welch t -tests (Bonferroni-corrected, $n_{\text{comparisons}} = 6$ per model) on first-token entropy. BW = Black Woman, BM = Black Man, WW = White Woman, WM = White Man. No White > Black contrast reaches Bonferroni-corrected significance in any model.

Model	Comparison	ΔH_1	d	p_{raw}	p_{Bonf}
Qwen3	BW–BM	+0.031	+0.028	.453	1.000
Qwen3	BW–WW	+0.027	+0.024	.528	1.000
Qwen3	BW–WM	+0.034	+0.031	.405	1.000
Qwen3	BM–WW	−0.005	−0.004	.909	1.000
Qwen3	BM–WM	+0.003	+0.003	.942	1.000
Qwen3	WW–WM	+0.008	+0.007	.851	1.000
Llama	BW–BM	−0.055	−0.057	.129	.772
Llama	BW–WW	+0.057	+0.056	.132	.790
Llama	BW–WM	−0.010	−0.010	.784	1.000
Llama	BM–WW	+0.112	+0.116	.002	.012*
Llama	BM–WM	+0.045	+0.049	.192	1.000
Llama	WW–WM	−0.066	−0.069	.064	.385
Falcon	BW–BM	−0.170	−0.162	< .001	< .001*
Falcon	BW–WW	+0.190	+0.170	< .001	< .001*
Falcon	BW–WM	+0.029	+0.027	.472	1.000
Falcon	BM–WW	+0.360	+0.334	< .001	< .001*
Falcon	BM–WM	+0.199	+0.193	< .001	< .001*
Falcon	WW–WM	−0.161	−0.147	< .001	.001*
Mistral	BW–BM	−0.044	−0.046	.213	1.000
Mistral	BW–WW	+0.056	+0.059	.112	.670
Mistral	BW–WM	+0.013	+0.014	.715	1.000
Mistral	BM–WW	+0.100	+0.107	.004	.025*
Mistral	BM–WM	+0.057	+0.061	.103	.617
Mistral	WW–WM	−0.044	−0.046	.214	1.000
Gemma	BW–BM	−0.013	−0.016	.673	1.000
Gemma	BW–WW	+0.009	+0.011	.773	1.000
Gemma	BW–WM	−0.001	−0.001	.982	1.000
Gemma	BM–WW	+0.022	+0.027	.472	1.000
Gemma	BM–WM	+0.012	+0.015	.680	1.000
Gemma	WW–WM	−0.010	−0.012	.749	1.000
OLMo	BW–BM	−0.124	−0.136	< .001	.002*
OLMo	BW–WW	+0.165	+0.182	< .001	< .001*
OLMo	BW–WM	+0.073	+0.081	.029	.175
OLMo	BM–WW	+0.289	+0.321	< .001	< .001*
OLMo	BM–WM	+0.197	+0.221	< .001	< .001*
OLMo	WW–WM	−0.092	−0.104	.005	.033*

Table 8: Full pairwise Welch t -tests on H_1 , Bonferroni corrected ($n_{\text{comparisons}} = 6$ per model). Stars: $p_{\text{Bonf}} < .05$. No White>Black contrast reaches significance in any model.

F Descriptive Statistics

Table 9 reports group means and standard deviations of first-token entropy (H_1). Black-associated groups (BW, BM) show numerically higher H_1 means than White-associated groups in all six models. Standard deviations are large relative to group means ($\text{SD}/M \approx 0.37\text{--}0.43$), consistent with wide per-prompt variability.

Model	Black Woman		Black Man		White Woman		White Man	
	M	SD	M	SD	M	SD	M	SD
Qwen3-8B-Base	3.916	1.109	3.885	1.131	3.890	1.143	3.882	1.108
Llama-3.1-8B	4.412	1.013	4.467	0.934	4.355	1.000	4.422	0.925
Falcon3-10B	4.164	1.088	4.334	1.014	3.974	1.137	4.135	1.052
Mistral-NeMo-12B	4.442	0.956	4.485	0.928	4.385	0.946	4.429	0.936
Gemma-2-9B	3.450	0.859	3.463	0.817	3.441	0.852	3.451	0.805
OLMo-2-7B	4.138	0.923	4.262	0.908	3.973	0.895	4.065	0.879

Table 9: Group means (M) and standard deviations (SD) of H_1 (nats), per model ($n = 1,440$ per cell). Black-associated groups show numerically higher H_1 in all six models.

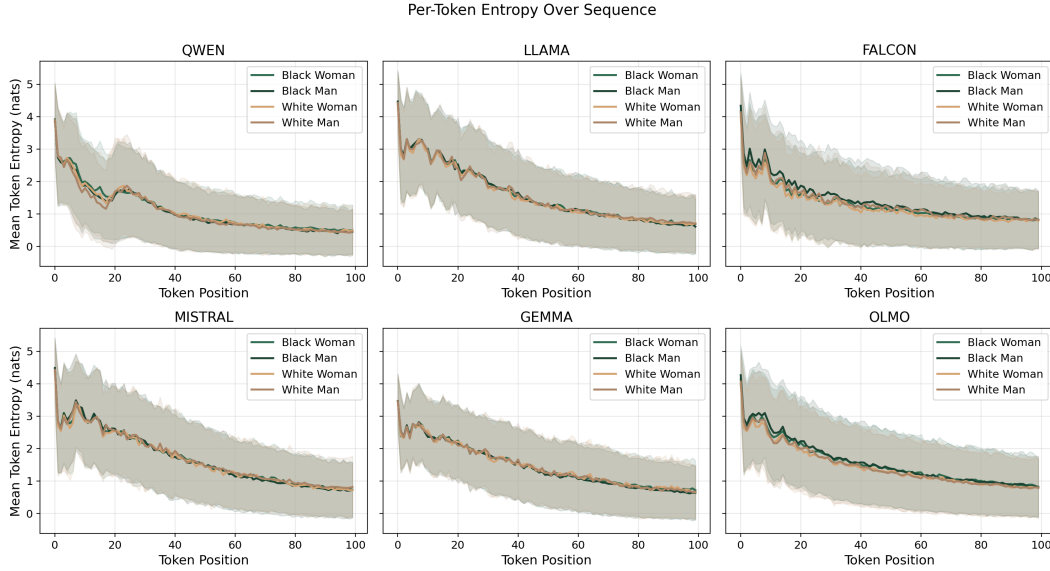


Figure 2: Mean per-token entropy (± 1 SD) over sequence position by demographic group and model. Group separation is most pronounced at early positions and decays as self-generated context accumulates.

G Mean-Entropy LME Race Effects

H Name Lists by Group

All names are drawn from Bertrand and Mullainathan [2004], Caliskan et al. [2017] and cross-referenced against Tzioumis [2018].

Black Woman: *Tanisha, Lakisha, Ebony, Shanice, Tamika, Keisha, Latoya, Shaniqua, Aaliyah, Imani, Nia, Amara, Zora, Destiny, Jasmine, Monique, Tiffany, Latasha, Shayla, Aisha, Brianna, Deja, Kiara, Leila, Nadia, Precious, Raven, Sasha, Tiana, Unique*

Black Man: *Jamal, DeShawn, Darnell, Tyrone, Marquis, Jermaine, Terrell, Darius, Devonte, Malik, Kwame, Kofi, Jabari, Rasheed, Lamar, Tremayne, Deandre, Antoine, Leroy, Maurice, Damon, Elijah, Isaiah, Jaylen, Jordan, Marcus, Nathaniel, Reginald, Shaun, Tyrell*

White Woman: *Emily, Claire, Megan, Stephanie, Katie, Lauren, Heather, Amber, Kristin, Molly, Hannah, Brittany, Allison, Sara, Rachel, Jessica, Ashley, Meredith, Courtney, Tara, Abigail, Alexandra, Caroline, Elizabeth, Grace, Isabella, Katherine, Natalie, Olivia, Samantha*

White Man: *Greg, Brett, Connor, Tyler, Cody, Jake, Kyle, Ryan, Brad, Chad, Derek, Evan, Luke, Nathan, Scott, Todd, Zach, Aaron, Andrew, Brandon, Christopher, Daniel, Ethan, Hunter, Jason, Jonathan, Matthew, Michael, Patrick, William*

Model	$\hat{\beta}_{\text{race}} (\bar{H})$	95% CI	p	sig
Qwen3-8B-Base	−0.013	[−0.065, +0.038]	.612	
Llama-3.1-8B	−0.003	[−0.043, +0.037]	.875	
Falcon3-10B	+0.140	[+0.081, +0.198]	< .001	*
Mistral-NeMo-12B	−0.030	[−0.057, −0.004]	.024	*
Gemma-2-9B	−0.011	[−0.035, +0.013]	.375	
OLMo-2-7B	+0.060	[+0.007, +0.112]	.027	*

Table 10: OLS (cluster-robust by name) race estimates for mean-over-sequence entropy ($\bar{H}$), controlling for tokenization length. Significant positive effects: Falcon3-10B and OLMo-2-7B. Mistral-NeMo-12B shows a significant *negative* effect (White > Black mean entropy). * $p < .05$.

Cross-demographic names: The following names have elevated cross-racial usage in Tzioumis [2018]: *Jasmine, Leila, Nadia, Sasha, Tiana, Brianna, Tiffany* (Black women) and *Jordan* (Black men). A restricted-pool sensitivity analysis excluding all eight is in Appendix I.

I Sensitivity Analysis: Restricted Name Pool

Table 11 compares race $\hat{\beta}$ under the primary model between the full 30-name pool and a restricted 22-name pool excluding the eight cross-demographic names.

Model	Full pool ($n = 5760$)		Restricted pool ($n = 4224$)		% change	Direction
	$\hat{\beta}_{\text{race}}$	p	$\hat{\beta}_{\text{race}}$	p		
Qwen3-8B-Base	+0.003	.929	+0.006	.876	+80.3%	stronger
Llama-3.1-8B	+0.045	.013	+0.043	.021	−5.1%	weaker
Falcon3-10B	+0.199	.000	+0.207	.000	+4.1%	stronger
Mistral-NeMo-12B	+0.057	.000	+0.055	.001	−2.6%	weaker
Gemma-2-9B	+0.013	.087	+0.010	.147	−16.6%	weaker
OLMo-2-7B	+0.197	.000	+0.198	.000	+0.3%	stronger

Table 11: Restricted-pool sensitivity: race $\hat{\beta}$ (primary model) for full vs. restricted name pool. Changes are mixed in direction and small in magnitude ($\leq 23\%$); all six race effects remain positive-directional.

I.1 White Man Pool Sensitivity

The White Man pool includes class-coded names (*Chad, Kyle, Brett, Brad, Connor, Cody, Jake*). Excluding these seven and re-estimating the frequency-controlled model:

Model	Full pool ($n_{\text{WM}} = 30$)		Restricted ($n_{\text{WM}} = 23$)		% change	Direction
	$\hat{\beta}_{\text{race}}$	p	$\hat{\beta}_{\text{race}}$	p		
Qwen3-8B-Base	+0.031	.466	+0.024	.601	−22.6%	weaker
Llama-3.1-8B	+0.052	.036	+0.061	.020	+16.7%	stronger
Falcon3-10B	+0.221	.000	+0.240	.000	+8.4%	stronger
Mistral-NeMo-12B	+0.039	.044	+0.048	.029	+21.4%	stronger
Gemma-2-9B	+0.016	.145	+0.017	.171	+4.9%	stronger
OLMo-2-7B	+0.159	.000	+0.165	.001	+3.9%	stronger

Table 12: White Man pool sensitivity under the frequency-controlled model. The race effect strengthens in 5/6 models after excluding class-coded names; the race effect is not an artifact of class-coding.

J Output Homogeneity: Implementation Details

Generated continuations are embedded with `all-mpnet-base-v2` Sentence-BERT [Reimers and Gurevych, 2019], with all occurrences of the generating name replaced by [NAME] (case-insensitive

whole-word regex) before encoding. In practice, 59–67% of continuations contain the generating name at least once, so this masking step is consequential for preventing name co-occurrence from artificially inflating within-group similarity.

For each (template t , group k) cell, the 30 name-conditioned continuations yield an embedding matrix $\mathbf{E} \in \mathbb{R}^{30 \times 768}$. Mean pairwise cosine similarity:

$$\text{MPCS}_{tk} = \binom{30}{2}^{-1} \sum_{i < j} \mathbf{e}_i^\top \mathbf{e}_j$$

Equation 3 is fit using statsmodels MixedLM with REML and lbfgs optimization. All six base models and six instruct models converge (convergence flag confirmed).

K Within-Prompt H_1 –Diversity Link

We test whether higher first-token entropy at the prompt level predicts more semantically diverse output within the same (template, group) cell. For each model:

$$d_{ij} = \gamma_0 + \gamma_1 H_{1,ij} + \gamma_2 \text{race}_i + \gamma_3 \text{gender}_i + u_j + \varepsilon_{ij} \quad (4)$$

where d_{ij} is cosine distance from the within-(template, group) centroid, u_j is a template random intercept, and race/gender covariates isolate the within-group relationship.

Model	Base models				Instruct models (chat)			
	$\hat{\gamma}_1$	95% CI	p	sig	$\hat{\gamma}_1$	95% CI	p	sig
Qwen	+0.001	[−0.011, +0.012]	.883		+0.121	[+0.088, +0.154]	< .001	***
Llama	−0.006	[−0.021, +0.011]	.502		+0.003	[−0.022, +0.028]	.804	
Falcon	+0.017	[+0.002, +0.031]	.026	*	+0.040	[+0.023, +0.056]	< .001	***
Mistral	+0.009	[−0.002, +0.019]	.104		+0.041	[+0.016, +0.067]	.001	**
Gemma	−0.004	[−0.020, +0.013]	.658		+0.038	[+0.023, +0.054]	< .001	***
OLMo	+0.006	[−0.006, +0.017]	.344		+0.021	[+0.006, +0.036]	.006	**
DL pooled	+0.005	[−0.001, +0.010]	.065		+0.042	[+0.020, +0.064]	< .001	***

Table 13: Within-prompt regression of output semantic distance on first-token entropy (Equation 4). For base models, the DL-pooled within-prompt relationship is near zero ($\hat{\gamma}_1 = +0.005$, $p = .065$), indicating that H_1 and MPCs are co-effects of demographic group membership rather than sequentially linked. For instruct models under chat formatting, the link is substantially stronger (DL-pooled $\hat{\gamma}_1 = +0.042$, $p < .001$; 5/6 significant). * $p < .05$, ** $p < .01$, *** $p < .001$.

For base models, the DL-pooled within-prompt $\hat{\gamma}_1 = +0.005$ ($p = .065$, $I^2 = 18.2\%$) indicates that higher H_1 at an individual prompt does not reliably predict more diverse output within the same template and group. The group-level co-occurrence of higher H_1 and more diverse outputs is more plausibly associated with a shared upstream factor (demographic group association) than with H_1 and output diversity being sequentially linked. For instruct models under chat formatting, 5/6 $\hat{\gamma}_1$ estimates are positive and significant ($I^2 = 86.2\%$), consistent with early distributional uncertainty being more tightly associated with output diversity under chat-formatted generation contexts.

L Frequency-Controlled Regression Estimates

The frequency covariate f_i is log-cumulative percent of births in US SSA records 1880–2008 [Wickham, 2014], mean-centered. White/Black names differ by a factor of 11.0 in mean cumulative prevalence (log ratio = 2.37 nats). The documented entropy asymmetry is the bias finding regardless of frequency control: underrepresentation in training data is itself a representational disparity, not a confound.

Model	Primary $\hat{\beta}_{\text{race}}$		+ntl _c		+ntl _c +freq+template	
	$\hat{\beta}$	<i>p</i>	$\hat{\beta}$	<i>p</i>	$\hat{\beta}$	<i>p</i>
Qwen3-8B-Base	+0.003	.929	+0.050	.238	+0.031	.466
Llama-3.1-8B	+0.045	.013	+0.079	.003	+0.052	.036*
Falcon3-10B	+0.199	.000	+0.255	.000	+0.221	.000*
Mistral-NeMo-12B	+0.057	.000	+0.061	.001	+0.039	.044*
Gemma-2-9B	+0.013	.087	+0.028	.003	+0.016	.145
OLMo-2-7B	+0.197	.000	+0.189	.000	+0.159	.000*

Table 14: Race $\hat{\beta}$ under three progressively controlled models for base models. Race effects persist in four of six models (Llama, Falcon, Mistral, OLMo) after frequency control. Stars (*): $p < .05$.

Model	Primary $\hat{\beta}_{\text{race}}$		+ntl _c		+ntl _c +freq+template	
	$\hat{\beta}$	<i>p</i>	$\hat{\beta}$	<i>p</i>	$\hat{\beta}$	<i>p</i>
Qwen3-8B	−0.007	.397	−0.014	.215	−0.023	.073
Llama-3.1-8B-I	+0.055	.000	+0.039	.019	+0.039	.029*
Falcon3-10B-I	−0.036	.574	−0.055	.516	−0.086	.410
Mistral-NeMo-I	−0.157	.005	−0.037	.499	+0.020	.740
Gemma-2-9B-IT	+0.012	.352	+0.011	.465	−0.011	.507
OLMo-2-7B-I	+0.065	.070	−0.004	.931	−0.042	.433

Table 15: Race $\hat{\beta}$ under three progressively controlled models for instruct models (chat-formatted probing). Only Llama retains significance after full control; Mistral’s large negative primary effect is absorbed by tokenization control.

M Interpretive Analysis: Sparsity vs. Contextual Associations

M.1 Sparsity Decomposition

Under the sparsity account, the cross-racial H_1 gap should scale with the cross-racial frequency gap via the within-race frequency slope. The race gap predicted under the frequency-indexed sparsity account is:

$$\widehat{\Delta H_1}^{\text{sparsity}} = \hat{\beta}_{\text{freq}} \times (-\Delta \log f)$$

Model	$\hat{\beta}_{\text{primary}}$	$\hat{\beta}_{\text{freq}}$	$\widehat{\Delta H_1}^{\text{sparsity}}$	% accounted for	Residual
Qwen3-8B-Base	+0.003	−0.010	+0.024	>100%	−0.021
Llama-3.1-8B	+0.045	−0.015	+0.035	77%	+0.010
Falcon3-10B	+0.199	−0.018	+0.043	22%	+0.156
Mistral-NeMo-12B	+0.057	−0.012	+0.028	49%	+0.029
Gemma-2-9B	+0.013	−0.007	+0.016	>100%	−0.003
OLMo-2-7B	+0.197	−0.017	+0.040	20%	+0.158

Table 16: Sparsity decomposition. In the two largest-effect models (Falcon3-10B, OLMo-2-7B), frequency-indexed sparsity accounts for only $\approx 20\%$ of the observed gap.

Within-group frequency slopes are consistently weak: Pearson r between per-name mean H_1 and log-prevalence is non-significant in 10 of 12 within-group regressions (Table 17). For Falcon3-10B specifically, the within-Black-name slope is $r = -0.010$ ($p = .94$), yet Falcon shows the largest cross-group gap.

M.2 Template-Category Heterogeneity of the Race Gap

Under pure frequency-indexed sparsity, the entropy elevation should be roughly constant across situational contexts. We test this by fitting a race $\times$ category interaction model with template fixed effects. The interaction is significant in all six base models (Qwen $F = 9.15$, $p < .001$; Llama $F = 5.54$, $p < .001$; Falcon $F = 3.26$, $p = .002$; Mistral $F = 8.16$, $p < .001$; Gemma $F = 12.58$,

Model	Black name pool			White name pool		
	slope	r	p	slope	r	p
Qwen3-8B-Base	−0.012	−0.107	.414	+0.002	+0.036	.785
Llama-3.1-8B	−0.011	−0.148	.261	−0.010	−0.247	.057
Falcon3-10B	−0.001	−0.010	.939	−0.027	−0.217	.096
Mistral-NeMo-12B	−0.010	−0.160	.222	−0.010	−0.292	.024*
Gemma-2-9B	−0.003	−0.095	.471	−0.003	−0.194	.138
OLMo-2-7B	−0.024	−0.210	.107	−0.007	−0.076	.563

Table 17: Within-group frequency- H_1 regression: per-name mean H_1 on log-prevalence, within Black ($n = 30$) and White ($n = 30$) name pools. Non-significant in 10 of 12 regressions. * $p < .05$.

$p < .001$; OLMo $F = 11.98$, $p < .001$). *Routine* and *Social* contexts show the largest race gaps (mean +0.138 and +0.111 nats across models); *Shopping* and *Neighborhood* contexts the smallest (+0.051 and +0.053 nats). Context-structured entropy gaps are inconsistent with uniformly flat priors from underrepresentation and more consistent with differences in the diversity of contextual co-occurrence patterns.

N Instruction-Tuned Models: Matched Raw-Text Probing

This appendix reports results for instruct models under matched raw-text probing (same inputs as base models, without chat templates). These results address the training-stage question; they do not characterize deployed behavior.

Model	ANOVA (H_1)		Race: Black–White		Gender: Woman–Man	
	$F(3, 5756)$	p	d [95% CI]	p	d [95% CI]	p
Qwen3-8B	7.43	< .001	+0.118 [±.052]	< .001	−0.037 [±.052]	.158
Llama-3.1-8B-I	9.23	< .001	+0.109 [±.052]	< .001	−0.085 [±.052]	.001
Falcon3-10B-I	19.54	< .001	+0.161 [±.052]	< .001	−0.121 [±.052]	< .001
Mistral-NeMo-I	7.13	< .001	+0.090 [±.052]	.001	−0.078 [±.052]	.003
Gemma-2-9B-IT	2.98	.030	+0.057 [±.052]	.031	−0.055 [±.052]	.039
OLMo-2-7B-I	16.24	< .001	+0.162 [±.052]	< .001	−0.082 [±.052]	.002

Table 18: One-way ANOVA and Cohen’s d for H_1 , instruct models under matched raw-text probing. All six race effects are positive and significant; DL random-effects pooled $\hat{\beta}_{\text{race}} = +0.153$ (95% CI [+0.091, +0.216], $p < .001$; $I^2 = 71.8\%$); $p_{\text{HMP}} = 1.5 \times 10^{-5}$ (one-sided). Compare with Table 20 (native chat probing), where the pattern is heterogeneous.

Model	Primary $\hat{\beta}_{\text{race}}$		+ntlc		+ntlc+freq+template	
	$\hat{\beta}$	p	$\hat{\beta}$	p	$\hat{\beta}$	p
Qwen3-8B	+0.181	.000	+0.261	.000	+0.233	.001*
Llama-3.1-8B-I	+0.109	.000	+0.150	.000	+0.099	.002*
Falcon3-10B-I	+0.187	.000	+0.260	.000	+0.222	.001*
Mistral-NeMo-I	+0.065	.022	+0.097	.002	+0.075	.027*
Gemma-2-9B-IT	+0.064	.003	+0.066	.008	+0.039	.130
OLMo-2-7B-I	+0.186	.000	+0.168	.003	+0.129	.017*

Table 19: Three-variant frequency-controlled model for instruct models under matched raw-text probing. Five of six models survive frequency control (only Gemma absorbed), contrasting with one of six under native chat-formatted probing (Table 15).

O Instruction-Tuned Models: Native Chat-Template Probing

Tables 20–22 report first-token entropy and output homogeneity results for instruct models under native chat-template formatting.

Model	ANOVA (H_1)		Race: Black–White		Gender: Woman–Man	
	$F(3, 5756)$	p	d [95% CI]	p	d [95% CI]	p
Qwen3-8B	2.62	.049	+0.035 [$\pm$.052]	.180	−0.013 [$\pm$.052]	.614
Llama-3.1-8B-I	2.26	.079	+0.034 [$\pm$.052]	.194	−0.003 [$\pm$.052]	.905
Falcon3-10B-I	16.31	< .001	+0.018 [$\pm$.052]	.487	+0.169 [$\pm$.052]	< .001
Mistral-NeMo-I	31.68	< .001	−0.161 [$\pm$.052]	< .001	−0.198 [$\pm$.052]	< .001
Gemma-2-9B-IT	2.44	.062	+0.049 [$\pm$.052]	.065	+0.045 [$\pm$.052]	.088
OLMo-2-7B-I	11.96	< .001	+0.103 [$\pm$.052]	< .001	−0.119 [$\pm$.052]	< .001

Table 20: One-way ANOVA and Cohen’s d for H_1 , instruct models under native chat formatting. The directionally uniform pattern of base models (Table 1) does not replicate; race d spans from negative (Mistral) to positive (OLMo).

Model	Race effect ($\hat{\beta}_{\text{race}}$, Black–White)				Gender ($\hat{\beta}_{\text{gender}}$, W–M)			
	$\hat{\beta}$	95% CI	p	sig	$\hat{\beta}$	95% CI	p	R^2
Qwen3-8B	−0.014	[−0.036, +0.008]	.215		−0.023	[−0.040, −0.006]	.009	.002
Llama-3.1-8B-I	+0.039	[+0.006, +0.071]	.019	*	+0.022	[−0.008, +0.053]	.158	.001
Falcon3-10B-I	−0.055	[−0.223, +0.112]	.516		+0.066	[−0.066, +0.198]	.325	.009
Mistral-NeMo-I	−0.037	[−0.145, +0.071]	.499		−0.093	[−0.195, +0.010]	.076	.025
Gemma-2-9B-IT	+0.011	[−0.019, +0.042]	.465		+0.010	[−0.014, +0.034]	.402	.001
OLMo-2-7B-I	−0.004	[−0.098, +0.090]	.931		−0.144	[−0.263, −0.025]	.017	.008

Table 21: OLS regression estimates (cluster-robust SEs) for H_1 , instruct models with native chat formatting, controlling for tokenization length. * $p < .05$.

P Generation Quality Audit

We report a systematic quality audit of all 69,120 generated continuations (5,760 records $\times$ 12 models).

Key findings from the audit: (1) zero real alignment refusals in any instruct model, confirming that raw-text implicit probing does not activate alignment suppression; (2) the OLMo-IT chat-leak race gap is +0.166 nats in the leak-excluded subset vs. +0.158 overall, confirming the leak does not attenuate the race result; (3) Gemma’s loop-free race gap is near zero (+0.001), confirming that Gemma’s small overall effect is partially loop-driven; (4) loop rates for instruct models are dramatically lower ($\leq 5.2\%$) than for the corresponding base models, consistent with RLHF/SFT co-occurring with fewer degenerate completions.

Model	Race: Black–White			Gender: Woman–Man		
	$\hat{\alpha}_{\text{race}}$	p	sig	$\hat{\alpha}_{\text{gender}}$	p	sig
Qwen3-8B	−0.036	< .001	***	+0.014	.121	
Llama-3.1-8B-I	−0.045	< .001	***	+0.030	.014	*
Falcon3-10B-I	−0.056	< .001	***	−0.014	.209	
Mistral-NeMo-I	−0.013	.334		+0.038	.006	**
Gemma-2-9B-IT	−0.044	< .001	***	+0.009	.258	
OLMo-2-7B-I	−0.035	< .001	***	−0.006	.446	

Table 22: LME MPCS results for instruct models under chat-formatted probing. Sign and direction consistent with base-model results (Table 4). * $p < .05$, ** $p < .01$, *** $p < .001$.

Model	Type	Mean ntok	%Max (100)	Chat leak	Real refusal	Loop ($\geq 3\times$)	EOS <30 tok
<i>Base models</i>							
Qwen3-8B-Base	base	86.0	72.5%	0	0	295 (5.1%)	10 (0.2%)
Llama-3.1-8B	base	98.2	95.5%	0	0	1,088 (18.9%)	24 (0.4%)
Falcon3-10B-Base	base	100.0	99.8%	0	0	149 (2.6%)	0
Mistral-NeMo-12B	base	100.0	99.9%	0	0	733 (12.7%)	0
Gemma-2-9B	base	99.9	99.6%	0	0	1,479 (25.7%)	0
OLMo-2-7B	base	93.5	86.0%	8 (0.1%)	0	366 (6.4%)	159 (2.8%)
<i>Instruction-tuned models (raw-text probing)</i>							
Qwen3-8B-Instruct	IT	100.0	100.0%	1 (<0.1%)	0	299 (5.2%)	0
Llama-3.1-8B-Instruct	IT	100.0	100.0%	0	0	15 (0.3%)	0
Falcon3-10B-Instruct	IT	96.5	90.7%	6 (0.1%)	0	14 (0.2%)	28 (0.5%)
Mistral-NeMo-Inst.	IT	97.7	92.9%	0	0	147 (2.6%)	24 (0.4%)
Gemma-2-9B-IT	IT	98.3	95.0%	0	0	1 (<0.1%)	6 (0.1%)
OLMo-2-7B-Instruct	IT	89.2	81.1%	60 (1.0%)	0	1 (<0.1%)	403 (7.0%)

Table 23: Generation quality diagnostics. Zero real alignment refusals across all 12 models. Principal anomalies: OLMo-IT health-template chat leaks (1.0%, race gap unaffected); repetition loops in three base models ($\leq 25.7\%$), with the race gap surviving loop-free subset analysis for Llama and Mistral; natural short completions from OLMo-IT (7.0%), which are coherent and contribute valid H_1 values.

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Answer: [Yes]

Justification: Twelve open-weight language models (six base, six instruction-tuned: Qwen3-8B-Base/Instruct, Llama-3.1-8B/Instruct, Falcon3-10B-Base/Instruct, Mistral-NeMo-12B/Instruct, Gemma-2-9B/IT, OLMo-2-7B/Instruct) are the subjects of study; their use is the core experimental method and is fully described in Section 3.3 and Table 1.

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